Population immunity dynamics under influenza–COVID-19 co-circulation reveal a lasting regime change

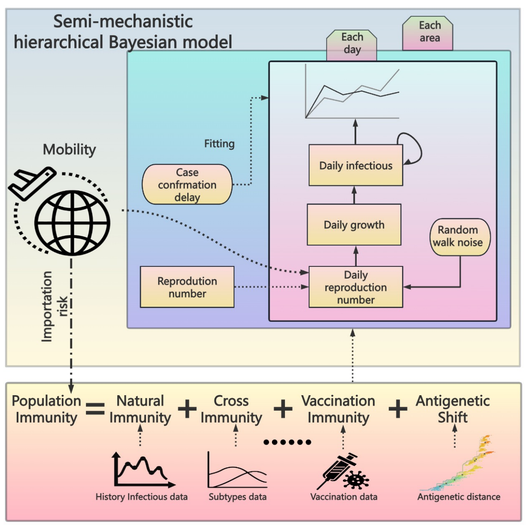
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Introduction

Whether COVID-19 merely interrupted seasonal influenza or durably altered its transmission ecology remains unclear. We reconstructed subtype-specific population immunity in Hong Kong from 2010 to 2025 using Bayesian models integrating syndromic, genomic, serological and international air-travel data. Influenza immunity showed subtype-specific sawtooth dynamics: epidemics emerged only after immunity returned to a lineage-stable floor, explaining up to 72% of within-subtype variation, whereas pooling subtypes obscured this relationship. Antigenic drift improved outbreak timing prediction, particularly for A(H1N1)pdm09 and A(H3N2). Joint influenza–COVID-19 modelling revealed persistent post-pandemic changes, including weakened viral interaction, restructured mainland-China importation, and reduced travel–influenza seasonality despite recovered mobility. These findings indicate a durable regime shift in influenza transmission ecology, challenging reliance on pre-pandemic baselines for forecasting and preparedness.

Methods



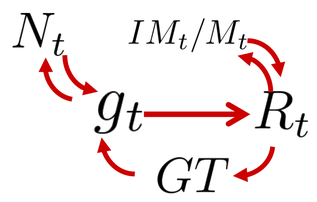
$$R_t^i = R_0^i \; exp\left(-\alpha_k IM^i(t) - \beta_j M(t)\right) \; exp\left(\Delta_t\right)$$

$$IM_{nat}^i(t) IM_{cro}^i(t) \; IM_{vac}^i(t) D^i[t,s] \;\; p_\infty\left(i|n_0\right)$$

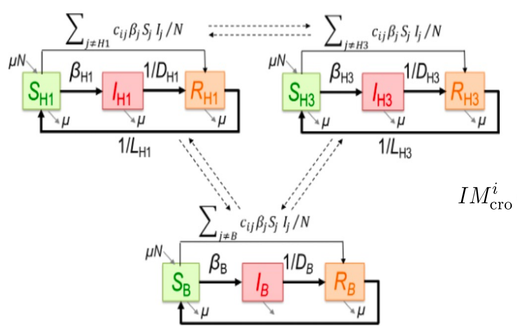
Transmission process

$$g_{t,c} = \exp\left(\beta_{\mathrm{GI}}\left(R_{c,t}^{\frac{1}{\alpha_{\mathrm{GI}}}} - 1\right)\right) - 1$$

$$N_{t,c}^{(C)} = N_{0,c}^{(C)} \prod_{\tau=1}^{t}\left[(g_{t,c}+1)\cdot \exp \epsilon_{t,c}^{(C)}\right]$$



Using a Comprehensive Dataset within a New Data-driven Sem-mechanistic Framework.



Mechanistic Model.

$$IM_{\mathrm{nat}}^i(t) = \sum_{s=1}^{m.\mathrm{lag}} im_{nat}(t-s)\, I^i(s) \exp\left(-\phi^i \times D^i[t,s]\right).$$

$$IM_{\mathrm{cro}}^i(t) = \sum_{j\neq i=1}^{n} \sum_{s=1}^{m.\mathrm{lag}} \left\{r_{ij} \times im_{\mathrm{cro}}(t-s)\, I^i(s) \exp\left(-\phi^i \times D^i[t,s]\right)\right\}.$$

Statistical Model.

Results

Figure 1. Time-varying population immunity to seasonal influenza subtypes inferred by three modelling frameworks, 2011–2020.

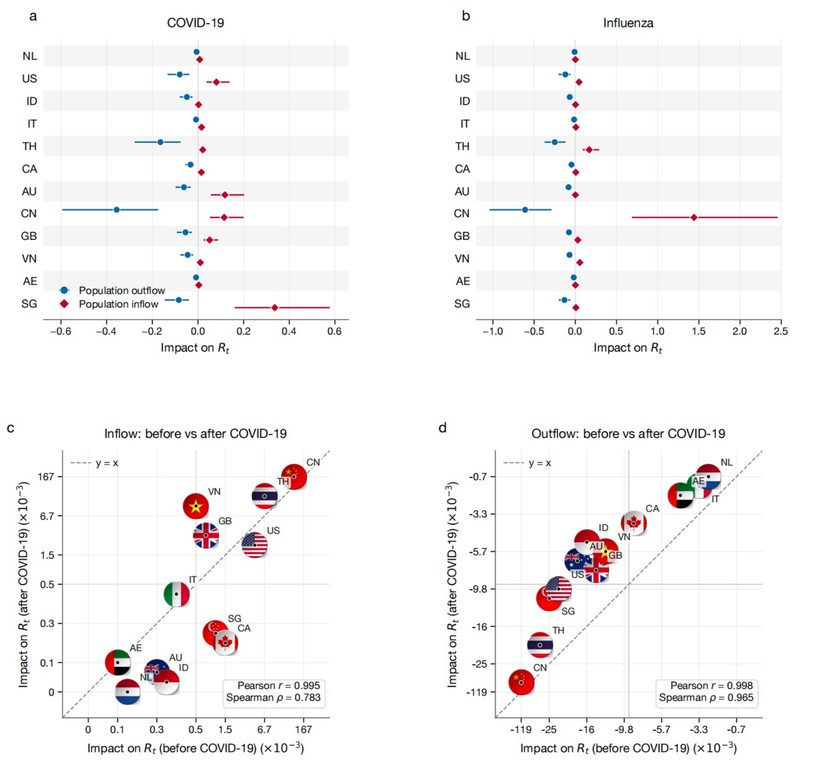


Figure 2. Effects of travel-associated risk from twelve source countries on the time-varying reproduction number (R_t) of COVID-19 and influenza in Hong Kong, with pre- versus post-pandemic comparison for influenza.

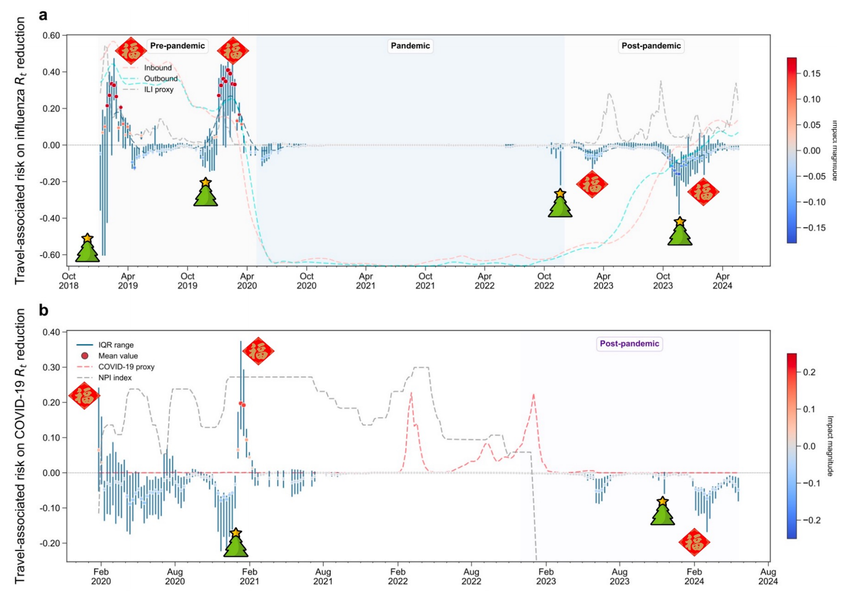


Figure 3. Impact of travel-associated risk on co-circulation before, during and after the pandemic.

Conclusion

- Time-varying immunity to seasonal influenza was inferred using statistical, semi-mechanistic, and mechanistic models.
- Antigenic factors are crucial for accurately predicting immunity.
- Inferred the Co-circulation dynamics and interactions (cross-immunity) between influenza and COVID-19 Before, During, and After the Pandemic.
- The pandemic reshaped the transmission patterns of seasonal influenza, and its patterns have not yet recovered fully in the pandemic period.

References

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[2] Chen Z, Tsui J L H, Gutierrez B, et al. COVID-19 pandemic interventions reshaped the global dispersal of seasonal influenza viruses[J]. Science, 2024, 386(6722): eadq3003.

Acknowledgement

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